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High-resolution regional gravity field modeling to connect the local vertical systems in South America to the IHRF

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Abstract

Since 2015, the scientific community has been working on unifying the global vertical system, following the resolution for the definition and realization of an International Height Reference System. This paper aims to determine the vertical datum parameter between the local vertical system and the global one in seven countries (Argentina, Brazil, Chile, Colombia, Ecuador, Venezuela, and Uruguay) in South America. For this purpose, Global Navigation Satellite System/leveling data and geoid undulations from a regional gravity field model were used in the computation. The results showed that the vertical datum parameter varies from 0.29 m (Talcahuano tide gauge—Chile) to 1.69 m (Santana tide gauge—Brazil). Besides that, geopotential numbers were estimated using the regional gravity field model computed in this paper and compared with the last version of the continental model. This research enhances geodetic knowledge by improving the accuracy of vertical data, utilizing advanced data correction methods, and integrating geoid models linked to international reference systems. The findings represent a significant advancement in addressing height inconsistencies, integrating national height systems within a global reference framework, and providing crucial data for diverse scientific and practical applications. These results have direct implications for infrastructure projects, natural hazard mitigation, and environmental monitoring in South America.

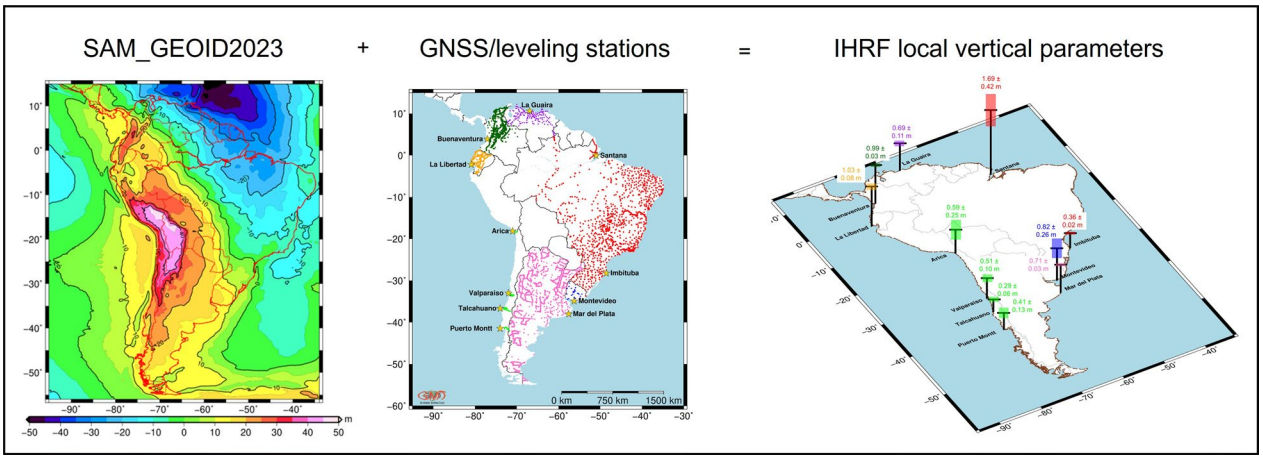
Keywords SIRGAS, South America vertical datum parameter, Geoid modeling, IHRF

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Graphical Abstract



1 Introduction

Historically, national height reference systems have been based on leveling networks, where height differences are determined through a combination of spirit leveling and gravimetry (not in all countries) (Grombein et al. 2017). A fixed datum point is chosen for each leveling network to establish the height reference level. In some countries, multiple datum points are selected (Ihde and Sánchez 2005). For practical purposes, the zero level of height systems is typically linked to the Mean Sea Level (MSL) realized by tide gauges, using sea level observations collected over a specific period to fix the zero level. However, due to variations in sea surface topography that depend on location and time, different tide gauges and observation periods may not establish the same height reference level. Consequently, national height systems that are generally linked to individual tide gauges rely on their Local Vertical Datum (LVD). The discrepancies between different LVD zones can lead to height datum offsets of approximately ± 1–2 m (Ihde and Sánchez 2005), making it challenging to directly compare the physical heights of different unconnected countries.

A classical height system consists of a series of stations established using geometric leveling, which are expressed in terms of height coordinates. The vertical system refers to the origin or zero point of this system, determined as the average local sea level measured by a tide gauge over a specific period. Due to the Earth's nutation movements, a minimum period of 18.6 years is required to establish a reliable average sea level value. However, not all countries have tide gauge measurements for this duration. In South America, height systems are based on MSL. Heights are determined through leveling circuits and lines measured during different epochs. These measurements have often been used without gravity measurements or adjustments applying various methodologies.

In South America, there are 18 height data points (Fig. 1), with 17 located in coastal areas and one on the continent (Paraguay) (Sánchez 2007). Some countries, such as Chile (which has six) and Argentina and Brazil (which each have two), maintain multiple height references (Sánchez 2007). In addition, the types of height



Fig. 1 Location of the vertical datum and types of heights in South America

are not homogeneous among South American countries. Brazil adopts normal heights (Fig. 1 in green), while Argentina, French Guyana, and Uruguay use orthometric heights (Fig. 1 in yellow). In Argentina, Mader orthometric heights have been derived from the computation of geopotential numbers obtained from a new adjustment of the vertical network (IGN-AR 2017). In Uruguay, Helmert orthometric heights are utilized. Bolivia, Chile, Colombia, Ecuador, Peru, and Venezuela employ what are known as normal-orthometric heights (Fig. 1 in red) (Sánchez 2007). There is no information available regarding the height system in the databases of the Geographical Institute for Paraguay, Guyana, and Suriname databases.

It is well-known that local height data are incompatible, because each reference point, such as a tide gauge, is situated on a different level surface. Several aspects contribute to this incompatibility:

- Gravimetric reductions, when applied, are often based on different formulations.
- Propagation errors arise from the leveling process.
- Time variations are not accounted for in height measurements and reference levels.

These three factors create uncertainties within local systems, making it difficult to exchange information between countries. According to Sánchez (2007), the differences between values measured simultaneously from different systems can reach up to ± 2 m globally. To illustrate the height discrepancies among South American countries, we collected height values from national agencies that have established leveling connections, resulting in 46 international connections (Fig. 2—blue dots). For instance, here is a height difference of up to 0.54 m between the Santana do Livramento station in Brazil and the Rivera station in Uruguay (Fig. 2) (IBGE 2019). In addition, the difference between the Monte Caseros station in Argentina and the Bella Unión station in Uruguay is 0.15 m (IGN-AR 2017). The difference between the Uruguaiana station in Brazil and the Paso de los Libres station in Argentina is 0.40 m (IBGE 2019).

From a regional perspective, integrating height information from multiple LVD is crucial for international engineering projects, flooding control, and exchange data which can present various challenges (Gerlach and Fecher 2012). Having a consistent physical height reference level is essential for accurately assessing global geodynamic and climatological processes within the Earth's system (Blewitt et al. 2010). This includes studies focused on monitoring global sea level rise and investigating coastal hazards. Establishing a consistent connection between geometric and physical reference frames is vital

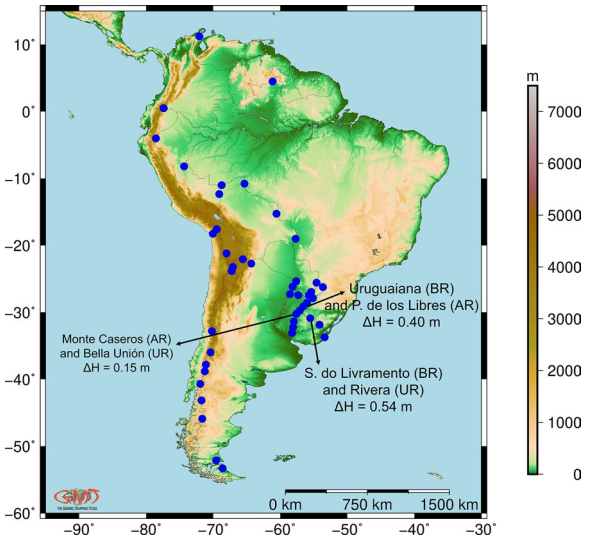


Fig. 2 Topography and leveling connections in South America (blue dots)

for geodetic applications such as unifying national gravity databases (Ihde and Sánchez 2005).

In 2015, during the International Union of Geodesy and Geophysics (IUGG) General Assembly, the International Association of Geodesy (IAG) released a resolution for the definition and realization of an International Height Reference System (IHRS) (Drewes et al. 2016). This system aims to standardize the vertical reference surface and establish a stable and consistent vertical component. According to Sánchez et al. (2016), the global equipotential reference surface was determined by the geopotential value of $W_0 = 62{,}636{,}853.4\ \mathrm{m^2 s^{-2}}$. Ihde et al. (2017), explained that the global vertical reference frame is a set of physical points materialized by coordinates (X_P) and distributed over the Earth's surface. The position P expressed in terms of its coordinates in the International Terrestrial Reference System (ITRS) is used to calculate geopotential number (C_P) and gravity potential (W_P) values concerning the geopotential (W_0). To determine the vertical components, it is necessary to calculate the potential differences (ΔW_P) between the Earth's gravity potential at point P and the reference potential of the vertical system (W_0). The potential difference $-\Delta W_P$ is designated as C_P. Based on the W_0 value, Sánchez and Sideris (2017) computed local gravity potential values at various tide gauges in South America and compared them with the previously established W_0. The vertical datum parameters estimated by Sánchez and Sideris (2017) in South America and Gómez et al. (2024) in Argentina were compared to the values obtained in this study.

This research aims to unify the International Height Reference Frame (IHRF) concerning seven existing

vertical systems in South America. First, we computed a regional gravity field model based on the Geodetic Boundary Value Problem (GBVP) solution utilizing terrestrial gravity, Global Geopotential Models (GGMs), and a Digital Elevation Model (DEM). We then combined the geoid model results with GNSS/leveling data to estimate the gravity potential of the zero-point height datum and each country's vertical datum parameter. In addition, to assess the quality of the gravity field model computed in this study (SAM_GEOID2023), we estimated geopotential numbers at the 17 IHRF points in South America and compared these with the most recent version of the continental model (SAM_GEOID2023).

In South America, several experiments related to the potential values at IHRF have been conducted (Tocho et al. 2020, 2022; Guimarães et al. 2021, 2022, 2024, 2023; Ribeiro et al. 2021; Silva et al. 2021, 2022; Gómez et al. 2024; Liu et al. 2024). However, the main challenge is determining the potential difference between the LVD and the global vertical datum (W0). Once this vertical datum parameter is determined, the LVD can be integrated into IHRS. Currently, there are two main strategies for vertical datum unification: the GBVP approach (i.e., geoid modeling) and the combination of GNSS/leveling data with either a GGM or a regional gravity field model (Amjadiparvar et al. 2016; Sánchez and Sideris 2017; Ebadi et al. 2019; Zhang et al. 2020, 2022a; Gómez et al. 2024). According to Wu et al. (2019) and Zhang et al. (2022b), the GBVP method is more rigorous in estimating the potential difference or vertical datum parameter between various vertical datum systems. However, it does have some drawbacks, including the use of different standards in various regions (such as gravity anomaly or gravity disturbance data) and limited access to terrestrial gravity data. The GBVP solution often relies on the remove–compute–restore (RCR) technique (Schwarz et al. 1990), which combines gravity data with a GGM and a DEM. Rangelova et al. (2015) determined the height datum unification in North America using the GBVP approach and tide gauge information. Ebadi et al. (2019) computed the Iranian height datum offset using GBVP solution and leveling/gravity data. Zhang et al. (2022b) refined the gravity field model using GBVP to compute China's height system unification. Liu et al. (2023) used the GBVP approach to determine the vertical datum parameters in Australia. The second strategy involves combining GNSS/leveling data with high-precision and high-resolution GGMs for vertical datum unification. Gomez et al. (2015) analyzed the discrepancies in vertical datum parameters between Argentina and Brazil. Tocho et al. (2022) computed the geopotential values in Argentina by integrating GNSS/leveling data and GGM. Grombein et al. (2017) utilized GOCE-based models to determine the vertical datum parameter in Austria, Brazil, and Germany. He et al. (2017) blended GRACE/GOCE-based models and GNSS/leveling data to assess the Chinese vertical datum. Kelly et al. (2021) computed the geopotential value of Ghana's height datum using the EIGEN-6C4 model and GNSS/trigonometric leveling data. Zhang et al. (2022a) estimated the height datum geopotential value of Hong Kong using GGMs and GNSS/leveling data.

2 Study area and data sets

This section briefly overviews the data sources utilized to compute the geoid model and estimate the vertical datum parameter. The data required consists of terrestrial gravity, global GGMs, DEM, and GNSS/leveling data.

2.1 Gravity data

Over the past 26 years, the Polytechnic School of the University of São Paulo (EPUSP) has developed a terrestrial gravity database for South America that includes 959,930 points (Fig. 3) (Blitzkow et al. 2016). Recently, the quality and availability of this important information have significantly improved across many South American countries.

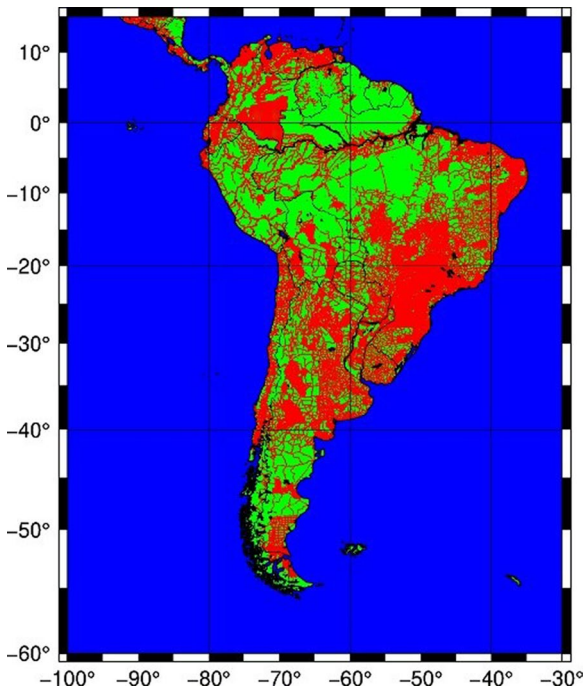


Fig. 3 Terrestrial gravity database in South America (in red). In green are the continental gaps in the geoid and quasi-geoid computations filled using the XGM2019e model up to degree and order 2190 (Zingerle et al. 2020). In blue, the multi-mission satellite altimetry model Grav_32.1 model (Available at: https://topex.ucsd.edu/pub/global_grav_1min) 5′ was used grid for the ocean area based (in blue—Fig. 3)

Furthermore, various universities and national mapping agencies have collaborated to conduct gravity measurements. It is important to note that the database excludes marine and airborne gravity data; only terrestrial gravity data were used for the geoid computation.

To compute geoid undulations, a considerable amount of gravimetric information is required. Therefore, the accuracy of gravity data is meticulously validated to ensure its quality. This validation process involved the use of DIVA software (Barriot and Sarrailh 2002), the SRTMv3 DEM model (Farr et al. 2007), and the EIGEN-6C4 (Förste et al. 2014) geopotential model up to degree and order 2190. This was accomplished by employing the gravity anomaly functional to validate the gravity data, height, and free-air gravity anomaly. To validate the data set using DIVA software, the free-air gravity anomaly (or Bouguer gravity anomaly) is predicted at the observed stations using the neighboring observations through a least squares collocation technique. This process results in a colored anomaly map based on triangulation. The user can then iteratively modify the map, removing and flagging any erroneous observations after careful control.

2.2 Global geopotential models

This study utilized four GGMs: EGM2008 (Pavlis et al. 2012), EIGEN6-C4 (Förste et al. 2014), XGM2019e (Zingerle et al. 2020), and SGG–UGM-2 (Liang et al. 2020). We selected the XGM2019e geopotential model to compute the geoid model, as it effectively represents both long- and medium-wavelength components using spectral decomposition and recomposition up to degree and order 300. In areas where gravimetric data were unavailable, we supplemented them with the XGM2019e gravity anomaly (up to degree and order 2190). The XGM2019e is a global gravitational field model that employs spheroidal harmonics functions up to degree and order 5399. It combines data from various sources, including the satellite model GOCO06s (Kvas et al. 2021) for long-wavelength components up to degree and order 300 and the terrestrial gravity grid for short wavelengths. Terrestrial data, which consists of both land and ocean gravity anomalies, was provided by the National Geospatial-Intelligence Agency at a resolution of 15' augmented with gravity information derived from the topographic model EARTH2014 (Hirt and Rexer 2015). In addition, DTU13 (Andersen et al. 2014), which provides satellite altimetry-derived gravity anomalies, was used for the oceans. The XGM2019e model was also used to compute the geopotential values at local vertical data and to validate the geoid and quasi-geoid models.

The GGMs EGM2008, EIGEN6-C4, and SGG–UGM-2 were used to estimate the geopotential values of a local vertical datum and to further validate the geoid and quasi-geoid models. The first was developed by combining data from the GRACE satellite, EIGEN-GL05C (Förste et al. 2008) of degree and order 360, and terrestrial and oceanic gravimetric data. The model is complete up to degree and order 2159 and contains additional coefficients up to degree 2190 and order 2159. In 2014, the GFZ/Potsdam and GRGS/Toulouse released the European Improved Gravity model of the Earth by New techniques known as EIGEN-6C4. This model incorporates data from various sources, such as the Laser Geodynamics Satellite (LAGEOS), Gravity Recovery and Climate Experiment (GRACE), Gravity Field and Steady-State Ocean Circulation Explorer (GOCE) (Rummel 2010) satellites, and the DTU12 satellite altimetry model (Andersen and Knudsen 2009). The coefficients of the series of spherical harmonic functions were computed up to degree and order 2190. The SGG–UGM-2 model was developed using the Ellipsoidal Harmonic Analysis and Coefficient Transformation theory. The model is complete up to degree 2190 and order 2159 and was created by combining observations from various sources, including GRACE and GOCE satellite data, marine gravity data from satellite altimetry, and continental gravity data derived from EGM2008.

2.3 Topographic data

The topographic data used in this study comes from the SRTMv3 model (Fig. 2; Farr et al. 2007), which has a resolution of 1″. This resolution was achieved by integrating data from the Shuttle Radar Topography Mission (SRTM) and the Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) mission for the continents.

To assess the impact of terrain effects on gravity anomalies, we utilized the TCGRID and TC programs from the GRAVSOFT package (Forsberg and Tscherning 2008). The methodology described by Forsberg (1984) involves quantifying topographic effects using a digital terrain model of the average elevation of the terrain. The reference DEM model was created by combining high-resolution and low-resolution. A coarse grid of 3' × 3', based on Vu et al. (2019), was generated from the detailed SRTMv3 model using the SELECT program from the GRAVSOFT package, employing a simple averaging method. Subsequently, the coarse model was filtered with a moving average operator, applying 16 sliding window factors with different resolutions. This process was iterative, with the statistics of the smoothed DEMs tested at each step, ultimately selecting the model with the smallest residual quantities.

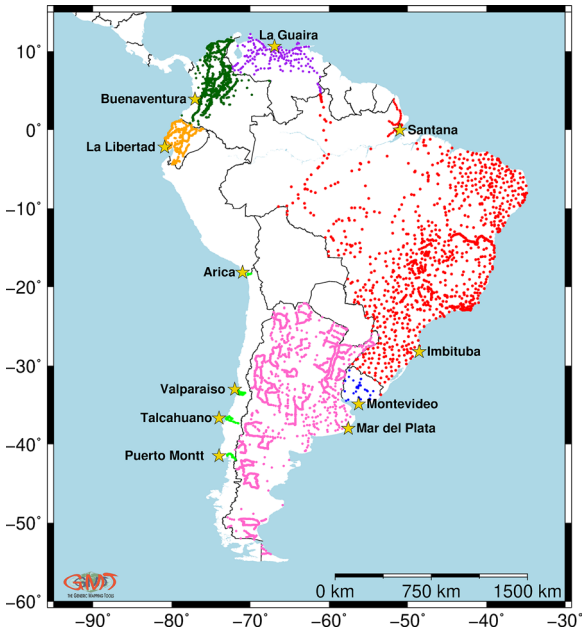


Fig. 4 Distribution of GNSS/leveling stations utilized in this study

2.4 GNSS/leveling data

The following institutions provided GNSS/leveling data benchmarks: the *Instituto Geográfico Nacional de Argentina* (IGN-AR), the *Instituto Brasileiro de Geografia e Estatística* (IBGE), the *Instituto Geográfico Militar de Chile* (IGM-Chile), the *Instituto Geográfico Agustín Codazzi* (IGAC), the *Instituto Geográfico Militar de Ecuador* (IGM-Ecuador), the *Instituto Geográfico Militar de Uruguay* (IGM-Uruguay), and the *Instituto Geográfico de Venezuela Simón Bolívar* (IGVSB) (Fig. 4). This figure illustrates the benchmarks distribution, with the following color coding: red dots in Brazil, blue dots in Uruguay, pink dots in Argentina, green dots in Chile, dark green dots in Colombia, orange dots in Ecuador, and purple dots in Venezuela. The leveling height values (usually in the mean tide concept) were transformed into the zero-tide concept according to the following equation (Ekman 1989; Tenzer et al. 2011):

$$H^{ZT} = H^{MT} + \frac{\Delta W_{ZT}}{g} \tag{1}$$

where H^{ZT} is height in the zero tide concept and H^{ZT} is height in the mean tide concept. $\Delta W_{ZT}/g$ can be computed with the sub-centimeter accuracy by

$$\frac{\Delta W_{ZT}}{g} \approx -0.198\left(\frac{3}{2}sin^2\varphi - \frac{1}{2}\right) \tag{2}$$

where ΔW_{ZT} is the gravitational potential in a zero tide concept, g is the observerd gravity, and \varphi is the latitude of the point.

The ellipsoidal height (in the free-tide concept) (h^{TF}) was transformed into the zero-tide concept using the following expression (Sánchez and Sideris 2017):

$$h^{ZT} = h^{TF} + 0.0602 - 0.1790sin^2\varphi - 0.0018sin^4\varphi \tag{3}$$

Leveling data in South America is heterogeneous. The leveling network in each country was established at different epochs, and the heights are based on different types (orthometric, normal, and normal-orthometric—see Table 1). In addition, information about height metadata is often lacking. Only Argentina[1] and Brazil[2] provide details on the precision of their leveling data, which is 6 cm for Argentina and between 1 and 15 cm for the Brazilian leveling network. In other cases, mapping agencies have no politics for open data. In this sense, in the American continent, SIRGAS (Geodetic Reference System for the Americas) has encouraged the national mapping agencies to open its repository and to share the information for educational and scientific purposes (see Resolution SIRGAS No. 1/2022) (SIRGAS 2022). A summary of height information used in this research is presented in Table 1.

The distribution of GNSS and leveling data in South America is uneven. The Amazon rainforest and the Andes mountains pose significant challenges in establishing new stations. In the Amazon, the lack of roads and paths makes it impossible to establish new stations. Similarly, the mountainous terrain of the Andes complicates this task. For instance, in Argentina and Ecuador, stations located above 4000 m were used in the study (Table 2). Regarding ellipsoidal height, Brazil and Venezuela have negative heights on the coast (up to − 20 m in Brazil and − 30 m in Venezuela) (Table 2). In terms of the separation between the ellipsoid and the geoid, it reaches 44.42 m in Argentina and 38.74 m in Venezuela. The amplitude reaches more than 55 m in Brazil while in Colombia around 46 m (Table 2).

3 Methods

In this section, we discuss the methodology of determining the geopotential value at the local data through the GBVP approach and estimating the vertical datum parameters.

[1] https://www.ign.gob.ar/NuestrasActividades/Geodesia/Nivelacion/Introduccion

[2] https://www.ibge.gov.br/geociencias/informacoes-sobre-posicionamento-geodesico/rede-geodesica/16283-rede-altimetrica.html?=&t=o-que-e

Table 1 GNSS/leveling data available for this study, vertical datum of each country, and type of height

Country	number of points	Institution responsible	Vertical Datum	Height characteristic	Website Data
Argentina	2931	IGN-AR	Mar del Plata	Orthometric height	Yes
Brazil	1350	IBGE	Imbituba (mainly) and Santana	Normal height	Yes
Chile	368	IGM-Chile	Arica, Talcahuano, Valparaíso (mainly), Puerto Montt	Normal-orthometric height	No
Colombia	5921	IGAC	Buenaventura	Normal-orthometric height	Yes
Ecuador	704	IGM-Ecuador	La Libertad	Normal-orthometric height	No
Uruguay	44	IGM-Uruguay	Montevideo	Orthometric height	No
Venezuela	190	IGVSB	La Guaira	Normal-orthometric height	No

3.1 Determination of the vertical datum geopotential value

The geopotential value (W_0^{LVD}) of a local vertical datum based on heterogeneous data measured at benchmark stations is estimated as (Kotsakis et al. 2012; Grigoriadis et al. 2017; Zhang et al. 2022a, b):

$$W_0^{LVD} = W_0 - \frac{\sum_{i=1}^{m}(h_i - H_i - N_I)\overline{\gamma}_i}{m}$$

$$= W_0 - \frac{\sum_{i=1}^{m}(N_i^{GNSS/lev} - N_i)\overline{\gamma}_i}{m} \quad (4)$$

where m is the number of GNSS/leveling benchmarks, W_0 is the global equipotential reference surface $W_0 = 62{,}636{,}853.4\ \mathrm{m^2 s^{-2}}$ (see Sánchez and Sideris 2017), h_i is the ellipsoidal height, H_i is the orthometric height, N_I is the geoid undulation (in this study, derived from a regional geoid model), $N_i^{GNSS/lev}$ is the geoid undulation derived from GNSS/leveling benchmarks, $\overline{\gamma}_i$ is the mean normal gravity on the reference ellipsoid computed through Somigliana's formula. For more details on the use of Eq. (4), see Jekeli (2000) and Kotsakis et al. (2012).

According to Kotsakis et al. (2012) the W_0^{LVD} uncertainty can be expressed as follows:

$$\sigma_{\widehat{W}_0^{LVD}} = \frac{\sqrt{\sum_i^m (H_i)^2}}{m}\sigma_g \quad (5)$$

where σ_g is the surface gravity accuracy.

The geoid undulation can be obtained by modeling the regional gravity field. The solution using GBVP provides a suitable approach for this modeling. The resolution of the GBVP to model the regional gravity field is usually treated by applying the integral Eq. (6). The integration considers the values of gravity acceleration, which are expressed in terms of gravity anomaly (Δg) (used in this study) or gravity disturbance (δg). The residual values (res) are calculated in both cases, with only the high-frequency being considered. The disturbing potential is defined as follows:

$$T = \frac{R}{4\pi}\iint_{\sigma} \Delta g_{res} S(\psi) d\sigma \quad (6)$$

where R is the mean radius of the Earth; Δg_{res} is the residual gravity anomaly and $S(\psi)$ is the Stokes' kernel function.

The geoid model computation using spectral decomposition integrates a global geopotential model with

Table 2 Statistics of the physical heights H', the ellipsoidal heights h, and the resulting height anomalies ($\zeta^{GNSS/lev}$) and geoid undulations ($N^{GNSS/lev}$) of the used GNSS/leveling benchmarks. All values are specified in [m]

Country	Type	Mean	Min	Max
Argentina	H'	632.494 ± 820.021	2.399	4859.332
	h	654.220 ± 825.741	17.773	4903.212
	$N^{GNSS/lev}$	21.731 ± 6.605	4.849	44.422
Brazil	H'	391.556 ± 316.434	1.917	1539.22
	h	370.709 ± 303.441	− 20.395	1523.794
	$\zeta^{GNSS/lev}$	− 8.850 ± 9.200	− 33.163	22.881
Chile	H'	183.792 ± 252.926	1.847	2006.883
	h	206.528 ± 255.839	21.332	2045.713
	$N^{GNSS/lev}$	22.735 ± 4.028	18.163	38.943
Colombia	H'	1049.035 ± 978.642	0.02	3849.588
	h	1062.707 ± 986.534	− 13.026	3865.266
	$N^{GNSS/lev}$	13.824 ± 11.956	− 15.137	31.749
Ecuador	H'	1256.714 ± 1241.568	2.376	4345.94
	h	1277.491 ± 1245.423	16.46	4373.778
	$N^{GNSS/lev}$	20.477 ± 4.860	10.334	29.243
Uruguay	H'	107.312 ± 66.553	5.15	285.28
	h	122.751 ± 66.806	17.82	300
	$N^{GNSS/lev}$	15.454 ± 1.160	12.393	17.547
Venezuela	H'	312.017 ± 383.621	0.751	2033.393
	h	294.242 ± 387.380	− 30.829	2016.942
	$N^{GNSS/lev}$	− 17.629 ± 8.561	− 38.742	1.982

terrestrial gravity observations and a digital terrain model around the area of interest. Stokes' formula requires gravity data covering the entire Earth. However, the global coverage of gravity data is incomplete. The computation relies solely on the available gravity data within and surrounding the region of interest to calculate the geoid undulation. However, performing residual Stokes integration over a limited area results in a truncation error, also known as the far-zone contribution of Stokes' integral, that affects the accuracy of the geoid undulation estimation. Many authors have investigated the theoretical basis for reducing the truncation error using modified integration kernels. These include deterministic approaches (Molodenskiĭ et al. 1962; Wong and Gore 1969; Meissl 1971; Vaníček and Kleusberg 1987; Heck 1988) and stochastic approaches (Wenzel 1982; Sjöberg 1984, 1991; Vaníček and Sjöberg 1991).

Regardless of the methodology chosen to obtain the geoid undulation values, a commonly used strategy is applying the remove–compute–restore procedure (Schwarz et al. 1990). In this sense, Eq. (6) considers the computation of the Stokes integral with the residual gravity anomalies, where the global geopotential model contribution is removed. To obtain high-frequency variability of the geoid, a DEM is applied in the Residual Terrain Model (RTM) process, to recover the geoid short-wavelength component. Then, the final residual gravity anomaly (Δg_{res}) is expressed as

$$\Delta g_{res} = \Delta g - \Delta g_{GGM} - \Delta g_{RTM} \tag{7}$$

where Δg is the gravity anomaly from observed gravity data, Δg_{GGM} is the gravity anomaly contribution from a Global Geopotential Model, and Δg_{RTM} is the gravity contribution of the Residual Terraim Model.

According to the generalized Burns' formula, the geoid undulation (N_P) at point P is expressed in terms of the disturbing potential and normal gravity ($N_P = T/\gamma$). By solving the GBVP, the result in terms of geoid undulation can be expressed as

$$N = N_{res} + N_{GGM} + N_{RTM} + N_0 \tag{8}$$

where N_{res} is the residual gravity field, N_{GGM} is the contribution of a GGM, N_{RTM} is the contribution of RTM and N_0 is the geoid zero-degree term; the contribution of the zero-degree term to the geoid undulation concerning a specific reference ellipsoid can be represented as (Heiskanen and Moritz 1967):

$$N_0 = \frac{GM - GM_0}{r_{P_0}\gamma_{Q_0}} - \frac{W_0 - U_0}{\gamma_{Q_0}} \tag{9}$$

where GM is the geocentric gravitational constant for a specific GGM ellipsoid, $GM_0 = 3{,}986{,}005 \times 10^8\ \mathrm{m^3\ s^{-2}}$, U_0 $= 62{,}636{,}860.85\ \mathrm{m^2\ s^{-2}}$ of the ellipsoid Geodetic Reference Ellipsoid 1980 (GRS80) (Moritz 2000), r_{P_0} is the geocentric radial distance and γ_{Q_0} is the normal gravity.

For the geoid (quasi-geoid) models computation, only the first term of Eq. (9) was used. The zero-degree term of − 0.93 m was added (*geoid undulation step* in Fig. 5).

3.1.1 Geoid undulation computation

To determine the vertical datum geopotential value, a regional geoid model is utilized, as illustrated in Fig. 5. This figure outlines the inputs, different steps, and modules of the SHGeo software package developed by the University of New Brunswick used in this process. The computation begins with the generation of complete Bouguer gravity anomalies from terrestrial gravity data and the simple Bouguer gravity anomalies from the XGM2019e model (degree 2190) with a 5' grid over the continental area. The GGM was used to fill the continental gaps (see Fig. 5—*input data step*). The simple Bouguer gravity anomalies were obtained from the International Centre for Global Earth Models (ICGEM) webpage (Ince et al. 2019). Subsequently, the SRTMv3 Digital Elevation Model was employed to calculate the terrain correction and add for obtaining the complete Bouguer gravity anomaly. These two data sets were then combined, and the Kriging interpolation method (using the GEOGRID program of the GRAVSOFT package) was applied to interpolate the point gravity anomaly data with a 1' resolution. After generating the 5' grid, the coordinates were placed at the center of the grid using the BLOCKMEAN subroutine from the GMT package (Wessel et al. 2019). In the next step, we utilized the FAIR program (SHGeo package) to obtain the mean free-air gravity anomalies from the mean complete Bouguer gravity anomalies.

The mean free-air gravity anomalies grid was completed using gravity anomalies from the Grav_32.1 model (Sandwell et al. 2014) over the ocean area; see Fig. 5 (combination step). This model has a 1' resolution, but it was transformed into a 5' grid using the SURFACE subroutine (GMT package), with the coordinates located at the center of the grid.

Stokes's method for solving the boundary value problem relies on gravity observations on the geoid as a boundary surface. On the other hand, gravity measurements are typically collected on the Earth's topography. To address this discrepancy, Helmert's second condensation method is applied. Helmert space is a harmonic space, i.e., no topographic mass above the geoid surface (Kingdon and Vaníček 2011). Therefore, the masses above the geoid are condensed into a layer at the geoid level. As a first step, various effects need to be added to the mean free air gravity anomaly to compute the mean Helmert gravity anomaly on the terrestrial surface. This

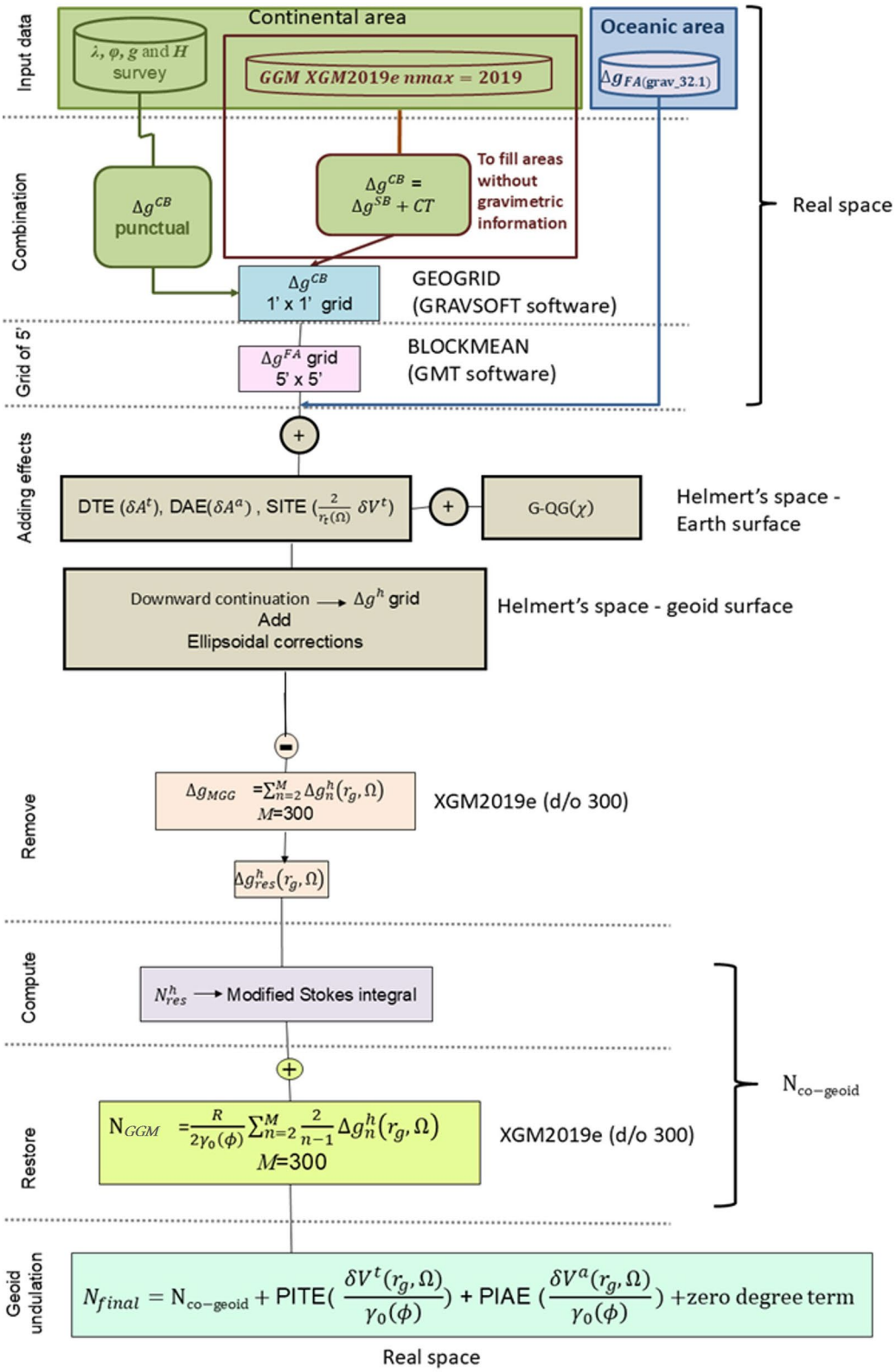


Fig. 5 Flowchart of sequential steps in the geoid undulation computation

process transforms the observed gravity anomaly on the Earth's surface (real space) into the Helmert gravity anomaly referred to the same surface in Helmert space. The topographic and atmospheric effects need to be formulated in their spherical form. The effects (referred to the Earth's surface) are presented below (see Ellmann and Vaníček 2007):

- Direct Topographical Effect (DTE) is the gravitational attraction of the condensed topographical masses (at the surface). It is subtracted from the attraction of real topographical masses (Martinec and Vaníček 1994a).
- Direct Atmospheric Effect (DAE) is similar to the previous one but only concerning atmospheric masses.
- The Secondary Indirect Topographical Effect (SITE) originates from applying the topographical correction in the gravity anomaly. It is computed as the difference between the topographical masses' gravitational potential and the condensation layer's gravitational potential (Vaníček et al. 2013).
- Geoid–quasigeoid correction (G–QG) is applied to the boundary condition formulated in Helmert space (Vaníček et al. 1999). This geoid model calculation used the orthometric height to calculate the punctual complete Bouguer gravity anomaly instead of the normal height. The geoid–quasigeoid correction is, therefore, the difference between the normal and orthometric heights, which can be approximated as a function of the simple Bouguer gravity anomaly (Martinec 1993).

In the sequence, the downward continuation (*adding effects step* in Fig. 5) of the mean gravity anomalies from the Earth onto the geoid was applied, where the boundary condition is established to solve the Stokes boundary value problem. Huang (2002) explained that "*The Downward Continuation (DC) can be considered as a 'projection' in which the gravity anomalies observed at a surface above the geoid are 'mapped' onto the geoid*". The DC problem is classified as an ill-posed problem, because a comparatively smooth function of the gravity anomaly is used to construct a more detailed and, therefore, 'rougher' function of the gravity anomaly (Huang 2002). However, Vaníček et al. (1996) demonstrated that the determination of the DC of mean Helmert's gravity anomalies on a grid of reasonable step (5') is a well-posed problem with a unique solution and can be done routinely to any accuracy desired in the geoid computation. After DC computation, the ellipsoidal correction for the gravity anomalies and the spherical approximation were applied to the Helmert gravity anomaly referred to the geoid's surface (Vaníček et al. 1999). Both were also used to compute two other effects added to the Helmert co-geoid (*geoid undulation step* in Fig. 5) to obtain the final geoid model (Ellmann and Vaníček 2007). The condensation of the atmospheric and topographic masses causes the Helmert potential to become slightly different from the actual potential. Consequently, the Helmert co-geoid does not exactly coincide with the geoid in real space. The effects causing this change are called (Ellmann and Vaníček 2007):

- Primary Indirect Topographic Effect (PITE) is computed from the gravitational potentials of the real and condensed topographical masses, both referred to the geoid level (Martinec and Vaníček 1994b).
- Primary Indirect Atmospheric Effect (PIAE) is similar to the previous one and concerns only the gravitational potentials of the real and condensed atmospheric masses.

The modified Stokes integral is a standard method for determining the short wavelength component of the geoid undulation. Geopotential models, which have developed since the Space Age, enable the separation of geoid undulation into long and medium wavelength components. The modified Stokes formula incorporates integration in a small circle around the computation point in the short-wavelength component, which is obtained (*compute step* in Fig. 5). The modified Stokes integral to obtain the geoid undulations (Ellmann and Vaníček 2007) is defined by the following equation:

$$N(\Omega) = \frac{R}{4\pi\gamma_0(\varphi)} \iint_{\Omega_{\psi 0}} \delta S_{vk}^{M}(\psi_0, \psi(\Omega,\Omega\prime)) \Delta g_{res}(r_g,\Omega) d\Omega\prime + \frac{R}{2\gamma_0(\varphi)} \sum_{n=2}^{M} \frac{2}{n-1} \Delta g_n^h(r_g,\Omega) + \frac{\delta V^t(r_g,\Omega)}{\gamma_0(\varphi)} + \frac{\delta V^a(r_g,\Omega)}{\gamma_0(\varphi)} \tag{10}$$

where $\delta S_{vk}^{M}(\Psi_0, \Psi(\Omega,\Omega\prime))$ is the modified Stokes function proposed by Vaníček and Kleusberg (1987) and defined as

$$\delta S_{vk}^{M}(\psi) = S^{M}(\psi) - \sum_{n=2}^{M-1} \frac{2n+1}{2} t_k(\psi_0) P_k(cos\psi). \tag{11}$$

where $S^{M}(\psi)$ is the original kernel, t_k are the coefficients determined from a system of linear equation, and $\Delta g_{res}(r_g,\Omega)$ is the residual Helmert gravity anomaly referred to the geoid's surface given by

$$\Delta g_{res}(r_g,\Omega) = \Delta g^h(r_g,\Omega) - \sum_{n=2}^{M} \Delta g_n^h(r_g,\Omega) \tag{12}$$

where R is the mean radius of the Earth; $\Psi(\Omega,\Omega\prime)$ is the spherical distance between the computation point and

the gravity data point; $d\Omega\prime$ is the integration area element. The geopotential model XGM2019e (d/o 300) was used as a reference to remove [second term on the right-hand side of Eq. (12)] (*remove step* in Fig. 5) the long and medium wavelength components of the Helmert gravity anomaly and after restoring [second term on the right-hand side of Eq. (10)] (*restore step* in Fig. 5) obtaining co-geoid, using the technique called "remove–compute–restore". The penultimate and final terms of Eq. (10) represent the PITE and PIAE (*geoid undulation step* in Fig. 5). The term $\Delta g^h\left(r_g, \Omega\right)$, on the right side of Eq. (12), is the Helmert gravity anomaly referring to the geoid's surface, which can be calculated using the Poisson integral equation (Heiskanen and Moritz 1967). It can be presented as

$$\Delta g^h(r_t, \Omega) = \frac{R}{4\pi r_t(\Omega)} \iint_{\Omega\prime\in\Omega_0} K[r_t(\Omega), \psi(\Omega, \Omega\prime), R]\Delta g^h\left(r_g, \Omega\prime\right) d\Omega\prime \tag{13}$$

where $K[r_t(\Omega), \psi(\Omega, \Omega\prime), R]$ is the spherical Poisson integral kernels (Sun and Vaníček 1998) and $\Delta g^h(r_t, \Omega)$ is the Helmert gravity anomaly referred to the Earth's surface. This last term is obtained by

$$\begin{aligned}\Delta g^h(r_t, \Omega) =& \Delta g^{FA}(r_t, \Omega) \\ &+ \delta A^t(r_t, \Omega) + \frac{2}{r_t(\Omega)}\delta V^t(r_t, \Omega) \\ &+ \delta A^a(r_t, \Omega) + \chi(r_t, \Omega)\end{aligned} \tag{14}$$

The first term on the right-hand side of Eq. (14) is the mean free air anomaly ($\Delta g^{FA}(r_t, \Omega)$); the second and third terms are the DTE and SITE; the fourth is DAE on gravitational attraction; and the last is G–QG. Figure 6 shows the mean free-air gravity anomaly on a 5' grid.

3.1.2 Geoid conversion to quasi-geoid

The conversion from geoidal to quasi-geoidal surfaces involves a correction related to the simple Bouguer gravity anomaly Δg_B and the height (H). The difference between the height anomaly (ζ) and the geoid undulation (N) is given by the following equation (Heiskanen and Moritz 1967):

$$\zeta = N - \frac{\Delta g_B H}{\gamma} \tag{15}$$

where γ is normal gravity. This transformation has some limitations once the Bouguer gravity anomaly depends on the mean elevation of the terrain, decreasing approximately by 0.1 Gal per 1 km average elevation (Hofmann-Wellenhof and Moritz 2006).

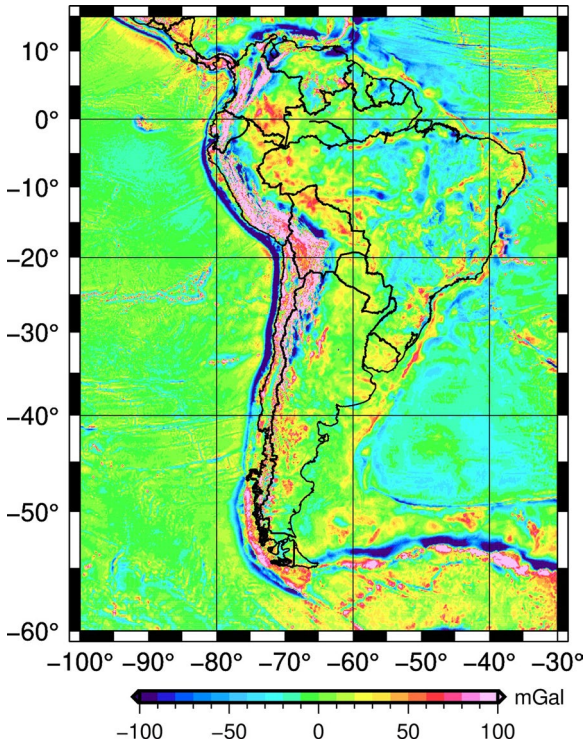


Fig. 6 Mean free air anomaly in South America

This conversion was performed by the N2ZETA program (Forsberg and Tscherning 2008). The routine calculates based on grids with a resolution of 5':

1. geoid undulation of the geoid model.
2. orthometric height generated from SRTMv3.
3. Bouguer gravity anomaly derived from the free air gravity anomaly grid.

The new geoid (quasi-geoid) model, named South AMerican GEOID 2023 (SAM_GEOID2023 and SAM_QGEOID2023), was developed in collaboration with SIRGAS WG-III. National mapping agencies across South America have been diligently working for many years to enhance the gravimetric data within the continent. The primary objective of this effort is to improve the geoid model consistently across South America. Recent updates to terrestrial gravity data have been made through new surveys conducted in Brazil, Argentina, Bolivia, Ecuador, and Venezuela.

The computation of the South American geoid began in 2010 with the SAM_GEOID2010 model (Matos et al. 2014). In 2015, the SAM_GEOID2015 model became available (Blitzkow et al. 2016). In 2021, we computed the SAM_GEOID2021 (Matos et al. 2021a) and SAM_QGEOID2021 (Matos et al. 2021b) models jointly with

Table 3 Characteristics of the four South American models

	SAM_GEOID2010	SAM_GEOID2015	SAM_GEOID2021	SAM_GEOID2023
Grid resolution	5′	5′	5′	5′
Number of gravity points	925,878	947,953	959,404	959,930
DEM	SAM3sv2	SAM3sv2	SRTMv3	SRTMv3
Gravimetric information gaps	Not completed	Not completed	XGM2019e (full degree)	XGM2019e (full degree)
GGM—remove and restore long and medium wavelengths	EGM2008 (n = m = 150)	EIGEN6-C4 (n = m = 200)	XGM2019e (n = m = 200)	XGM2019e (n = m = 300)
Software package	SHGeo	SHGeo	SHGeo	SHGeo
Downward continuation	No	No	No	Yes
Integral	Modified Stokes	Modified Stokes	Modified Stokes	Modified Stokes
Modified Stokes function	Molodenskii-Meissl	Molodenskii-Meissl	Vaníček and Kleusberg	Vaníček and Kleusberg
Permanent Tide	Tide-free	Tide-free	Zero-tide	Zero-tide
Zero-degree term [m]	− 0.41	− 0.41	− 0.17	− 0.93
W_0 [m^2/s^2]	62,636,855.69	62,636,855.69	62,636,853.40	62,636,853.40

Working Group III of SIRGAS (Geodetic Reference System for the Americas) (see https://sirgas.ipgh.org/en/organization/working-groups/working-group-iii/). All these models are available on the IAG International Service for the Geoid (ISG) website. Over the years, the amount of gravity data and the enhancement of the global models (especially the geopotential models derived from the gravity mission) contributed to the improvement of the recent geoid models. Table 3 summarizes the four geoid models, highlighting their differences in available data, GGM, and methods.

3.2 Estimation of the local vertical datum parameters

An equipotential surface defines the global vertical datum within the IHRS, with its potential value of $W_0 = 62{,}636{,}853.4\ m^2s^{-2}$. The level difference between the global and local vertical data ($W_0 - W_{0i}$), ensuring that the W_{0i} values are not arbitrary but consistent with the global W_0 is known as "vertical datum parameters".

The GBVP approach presented in Sect. 3.1. Equation (6) has a direct bias term caused using biased boundary values (gravity anomalies, geopotential numbers, etc.), which are referenced to different local vertical data W_{0i} (Sánchez and Sideris 2017). This implies that the difference (ΔW_0) between the potential at a point P (W_P) on the surface and the potential at a point Q (U_Q) in the reference ellipsoid (where the point Q is the projection of the surface point P onto the telluroid along the ellipsoid's normal) can have as many values as there are existing W_{0i} data. On the other hand, each datum can be split into a common component applicable to all vertical data ΔW_0 and a complementary component δW_{0i}, which illustrates the relationship between the common level W_0 and the individual levels W_{0i} (Sánchez and Sideris 2017). The individual level corresponds to the vertical datum offset. For a specific point P, the vertical datum offset (δH_P), can be expressed as (Zhang et al. 2020)

$$\delta H_P = h_P - H_P - N_0 - N^P_{GGM} - N^P_{RTM} - N^P_{res} \quad (16)$$

It is worth mentioning that for the vertical datum offset computation the second term of Eq. (9), (zero-degree term (N_0)) was considered. Equation (16) can be summarized as follows:

$$\delta H_P = h_P - H_P - N_P \quad (17)$$

where h_P is the ellipsoidal height, H_P is the orthometric height, and N_P is estimated according to Eq. (9).

According to Eq. (16), the vertical datum offset δH_P should ideally be a constant value. However, there may be slight variations in vertical datum parameters due to systematic errors and distortions in the leveling network and random errors in the computation of geodetic and geoid undulations. To minimize both random and systematic errors, a parametric model (Hayden et al. 2012; Grombein et al. 2017; Zhang et al. 2020) can be utilized to estimate the vertical datum parameter. The observation equation can be formulated as

$$\begin{aligned} l_P =& h_P - H_P - N_P = \delta H + x_1(\varphi_P - \varphi_0) \\ &+ x_2(\lambda_P - \lambda_0)cos\varphi_P \end{aligned} \quad (18)$$

where δH is the mean vertical datum offset, (φ_P, λ_P) are the latitude and longitude of the GNSS/leveling benchmarks, (φ_0, λ_0) are the mean latitude and longitude of all GNSS/leveling benchmarks, x_1 is the north–south tilt parameter, and x_2 is the east–west tilt parameter.

The functional model, according to Eq. (18), can be represented in matrix form as

$$\underbrace{\begin{pmatrix} l_1 \\ l_2 \\ \vdots \\ l_i \end{pmatrix}}_{l} = \underbrace{\begin{pmatrix} 1 & (\varphi_1-\varphi_0) & (\lambda_1-\lambda_0)cos\varphi_1 \\ 1 & (\varphi_2-\varphi_0) & (\lambda_2-\lambda_0)cos\varphi_2 \\ \vdots & \vdots & \vdots \\ 1 & (\varphi_i-\varphi_0) & (\lambda_i-\lambda_0)cos\varphi_i \end{pmatrix}}_{A} \cdot \underbrace{\begin{pmatrix} \delta H \\ x_1 \\ x_2 \end{pmatrix}}_{x} \quad (19)$$

where l is the observation vector, A the design matrix containing the observation's partial derivatives and x is the unknown vector. The system solution is possible using the Least Square Adjustment where the vertical datum offset, and the tilts are estimated by

$$\widehat{x} = (A^T PA)^{-1} A^T Pl \quad (20)$$

where P is the weight matrix. This paper assumes $P=I$, where I is the identity weight matrix.

The local vertical datum parameter (i.e., the mean vertical offset) is related to the potential difference between the reference potential W_0 and the local W_0^{LVD} and is formulated as follows

$$\delta H = \frac{\delta W}{\overline{\gamma}} = \frac{W_0 - W_0^{LVD}}{\overline{\gamma}} \quad (21)$$

4 Results and discussion

4.1 Geoid and quasi-geoid model

The geoid model of South America (Fig. 7a) was developed as outlined in Sect. 3.1.1, utilizing the flowchart presented in Fig. 5. Considering only the region in the continent, the maximum and minimum values in terms of geoid undulations are 50.57 m (in the Bolivian Andes) and − 44.99 m (in the Venezuelan coast). The differences between the geoid and the quasi-geoid models (Fig. 7b) are more evident in the Andes region than the other regions. The minimum difference of − 2.48 m is observed in the Argentinean Andes, while the maximum difference of 1.57 m occurs in northern Colombia.

4.1.1 Validation of geoid/quasi-geoid modeling with GNSS/leveling data

SAM_GEOID2023 and SAM_QGEOID2023 were evaluated using geoid undulations and height anomalies referenced from GNSS/leveling points. These points were obtained by calculating the differences between normal-orthometric (Chile, Colombia, Ecuador, Venezuela) or orthometric (Argentina and Uruguay) or normal (Brazil) heights (Table 4). Initially, these heights are based on a mean-tide concept and originate from leveling points and

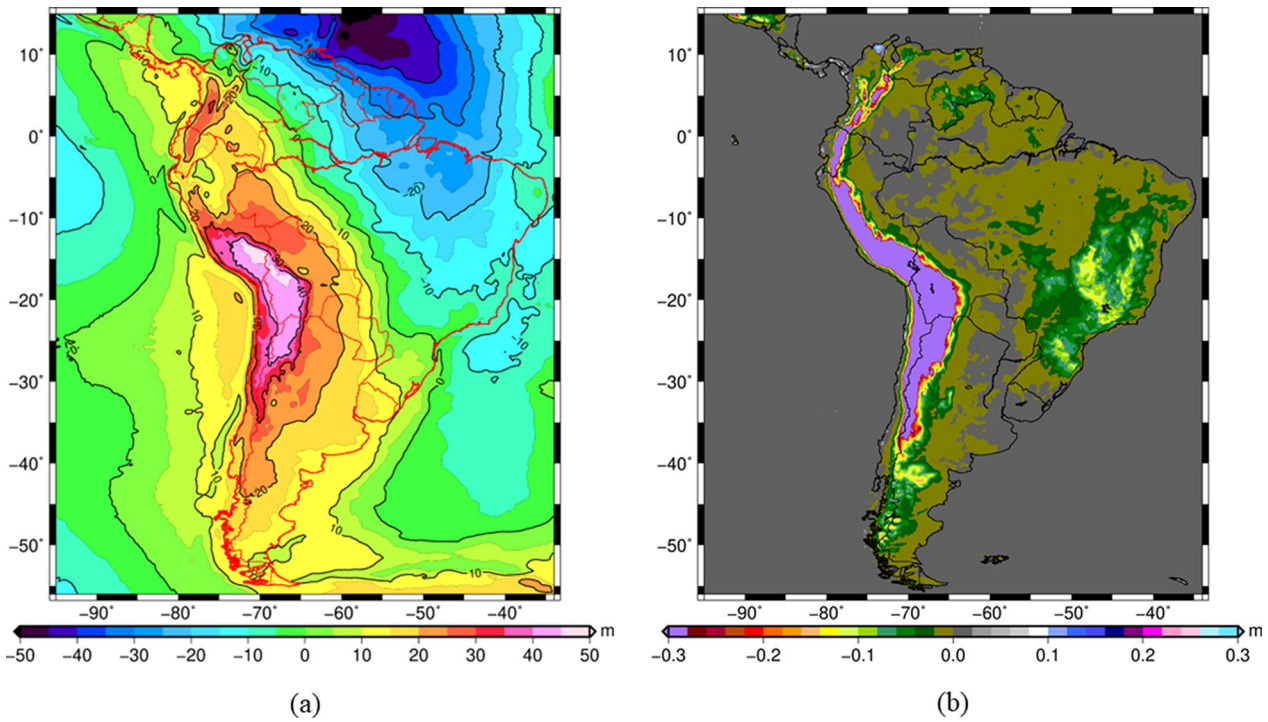


Fig. 7 **a** South American geoid model 2023 (SAM_GEOID2023). **b** Difference between the geoid (SAM_GEOID2023) and the quasi-geoid (SAM_QGEOID2023) models

Table 4 Summary of absolute statistical analysis of standard deviation and (mean) (in meters). Geoid undulations difference among GNSS/leveling data and SAM_GEOID2023 and GGMs (d/o 2190) for Argentina, Chile, Colombia, Ecuador, Uruguay, and Venezuela; height anomalies difference among GNSS/leveling data SAM_QGEOID2023 and GGMs (d/o 2190) for Brazil

Country	EGM2008 (2190)	EIGEN6C4 (2190)	XGM2019e (2190)	SGG-UGM-2 (2190)	SAM_GEOID2023 SAM_QGEOID2023
Argentina (2,922 points)	0.28 (0.76)	0.27 (0.76)	0.26 (0.75)	0.26 (0.76)	0.25 (0.72)
Brazil—Imbituba (1,271 points)	0.27 (0.41)	0.21 (0.42)	0.19 (0.42)	0.21 (0.43)	0.18 (0.44)
Brazil—Santana (66 points)	0.21 (1.77)	0.11 (1.76)	0.10 (1.74)	0.11 (1.77)	0.08 (1.78)
Chile-Arica (27 points)	0.28 (0.94)	0.33 (0.77)	0.29 (0.67)	0.28 (0.80)	0.17 (0.66)
Chile-Valparaíso (127 points)	0.24 (0.80)	0.20 (0.46)	0.20 (0.55)	0.22 (0.53)	0.19 (0.52)
Chile-Talcahuano (118 points)	0.18 (0.18)	0.16 (0.29)	0.14 (0.32)	0.16 (0.32)	0.16 (0.29)
Chile-Puerto Montt (84 points)	0.29 (0.70)	0.34 (0.69)	0.34 (0.69)	0.35 (0.71)	0.34 (0.78)
Colombia (5,837 points)	0.45 (1.10)	0.40 (1.13)	0.38 (1.09)	0.40 (1.13)	0.34 (0.92)
Ecuador (692 points)	0.47 (1.01)	0.43 (1.03)	0.44 (0.99)	0.44 (1.02)	0.39 (0.86)
Uruguay (44 points)	0.09 (0.80)	0.09 (0.81)	0.08 (0.79)	0.08 (0.79)	0.08 (0.83)
Venezuela (180 points)	0.23 (0.79)	0.21 (0.79)	0.20 (0.78)	0.21 (0.79)	0.19 (0.78)

their corresponding ellipsoidal heights (which are in tide-free concept). Therefore, these heights were transformed to zero-tide Eqs. (1) and (3) to obtain geoid undulations and height anomalies in the same tidal concept of the models computed and GGMs (EGM2008, EIGEN6C4, XGM2019e, and SGG–UGM-2). For validation in absolute terms, linear interpolation was used to obtain the geoid undulations and height anomalies from the models (SAM_GEOID2023 and SAM_QGEOID2023) using the coordinates of the GNSS/leveling points. For the GGMs, the computation service from the ICGEM website was used (Ince et al. 2019). Geoid and height anomaly functional in "User-defined point coordinates" were chosen, the reference system was GRS80, the tide concept was zero-tide, and the zero-degree term was added for these coordinates. The ICGEM considers $W_0 = U_0$ by Eq. (9) too.

The SAM_GEOID2023 and SAM_QGEOID2023 present a standard deviation compared to GNSS/leveling from 0.08 m (Uruguay) to 0.39 m (Ecuador). Except in Chile–Puerto Montt, Chile–Talcahuano, and Uruguay, in all countries, the geoid (quasi-geoid) performed better than the GGMs compared to the GNSS/leveling stations. Notably, most of the gravity data used for the geoid computation were also utilized in the GGMs computation, particularly XGM2019e (Zingerle et al. 2020), where the results are slightly worse than the geoid (quasi-geoid) models. Furthermore, XGM2019e (full degree) was employed in geoid modeling in areas lacking gravity measurements. For instance, in

Table 5 Summary of absolute statistical analysis of standard deviation and (mean) (in meters) involving the South American geoid models (SAM_GEOID2010, SAM_GEOID2015, SAM_GEOID2021 and SAM_GEOID2023) and quasi-geoid models (SAM_QGEOID2021 and SAM_QGEOID2023)

Country	SAM_GEOID2010	SAM_GEOID2015	SAM_GEOID2021 SAM_QGEOID2021	SAM_GEOID2023 SAM_QGEOID2023
Argentina (2,922 points)	0.44 (-0.87)	0.38 (-0.71)	0.31 (-0.02)	0.25 (0.72)
Brazil - Imbituba (1,271 points)	0.33 (0.20)	0.28 (0.16)	0.22 (-0.26)	0.18 (0.44)
Brazil - Santana (66 points)	0.16 (-0.66)	0.12 (-0.83)	0.07 (1.02)	0.08 (1.78)
Chile-Arica (27 points)	0.29 (-0.19)	0.27 (0.18)	0.14 (-0.18)	0.17 (0.66)
Chile-Valparaíso (127 points)	0.52 (-0.44)	0.34 (-0.29)	0.25 (-0.03)	0.19 (0.52)
Chile-Talcahuano (118 points)	0.32 (-0.11)	0.18 (-0.15)	0.16 (-0.40)	0.16 (0.29)
Chile-Puerto Montt (84 points)	0.38 (-0.61)	0.30 (-0.53)	0.62 (0.29)	0.34 (0.78)
Colombia (5,837 points)	0.84 (-1.22)	0.74 (-0.87)	0.43 (0.32)	0.34 (0.92)
Ecuador (692 points)	0.89 (-1.12)	0.81 (-0.79)	0.51 (0.39)	0.39 (0.86)
Uruguay (44 points)	0.15 (-0.87)	0.17 (-0.77)	0.09 (0.46)	0.08 (0.83)
Venezuela (180 points)	0.47 (-0.02)	0.42 (0.11)	0.26 (0.17)	0.19 (0.78)

Argentina, nearly 30% of GNSS/leveling comes from regions without gravity data which helps explain the similar results between the geoid model and the GGMs.

The models presented in this paper were also compared with the oldest South American models SAM_GEOID2010, SAM_GEOID2015, SAM_GEOID2021 and SAM_QGEOID2021 (Table 5). In Argentina, the Standard Deviation (STD) for SAM_GEOID2023 was reduced by 21% compared to SAM_GEOID2021 and by 43% compared to SAM_GEOID2010. In Brazil, the STD related to GNSS/leveling data linked to Imbituba tide gauge, decreased 46% between SAM_GEOID2010 and SAM_QGEOID2023 models. In Chile, the STD mean across all tide gauges decreased 42% from SAM_GEOID2010 to SAM_GEOID2023. Comparing the 2010 and 2023 models, improvements of approximately 60% and 57% were observed in Colombia and Ecuador, respectively. In Uruguay, the STD decreased 50%, while in Venezuela 60% considering the models from 2010 and 2023.

4.1.2 Adjustment of combined geoid and GNSS/leveling networks

According to Fotopoulos et al. (1999), the ideal equality $N=h–H$ is often unmet due to various factors. These factors include (1) inconsistencies in data, which refers to the discrepancies between the gravimetric quasi-geoid and the reference surface of the leveling datum; (2) random noise and errors, as well as systematic distortions in both ellipsoidal and physical heights; and (3) geodynamic effects, such as surface deformation caused by earthquakes, particularly when GNSS and leveling data are collected over an extended period. To model these effects, a correction surface was applied to the GNSS/leveling-based heights using the following basic model (Fotopoulos et al. 1999):

$$h_i - H_i - N_i = a_i^T x + v_i \tag{22}$$

where x is a vector of n unknown parameters, a_i is the coefficients vector and H_i denotes a residual random noise term. For this study, the four-parameter model was used:

$$a_i^T x = x_0 + x_1 cosx\varphi_i cos\lambda_i + x_2 cos\varphi_i sin\lambda_i + x_3 sin\varphi_i \tag{23}$$

The correction was applied to identify a more reliable accuracy assessment for the geoid (quasi-geoid) model once the national agencies provided the GNSS/leveling data and did not provide metadata information. The comparison between the calculated models and GNSS/leveling data before and after applying the correction surface is shown in Table 6. The standard deviation value of the differences decreased after applying the correction, especially in Brazil–Imbituba (before 0.18 m and after 0.12 m), Chile–Arica (before 0.17 m and after 0.09 m), Chile–Puerto Montt (before 0.34 m and after 0.20 m).

Even after applying the surface correction, many GNSS/leveling stations (Fig. 8) present discrepancies concerning the geoid (quasi-geoid) models. Most of these stations are located in mountainous regions. These discrepancies are expected due to the region's challenging topography and the gaps in terms of gravity data. In the areas, where data is available, terrestrial gravity, and leveling probably were collected along roads, which typically follow the slopes of the mountains. For the stations in flatter areas, we recommend that the national agencies check those stations.

An analysis (Fig. 9) was performed to assess the differences: ΔN represents the difference in geoid undulations between the geoid model and GNSS/leveling data, while $\Delta\zeta$ denotes the difference in height anomalies between the quasi-geoid model and GNSS/leveling data. Both ΔN and $\Delta\zeta$ are plotted as blue dots on the left side of Fig. 9, with dashed yellow lines indicating the location of the vertical data for each country (Brazil and Chile have more than one local vertical datum). Based on Fig. 9 (left-hand side), it is not possible to confirm that ΔN and/or $\Delta\zeta$ increase as they move away from the LVD. On the other hand, Fig. 9 (red dots on the right-hand side) shows that ΔN and $\Delta\zeta$ increase in the higher stations, especially in Argentina and Ecuador. Those points are in the Andes regions, contributing to an inaccuracy of that region's leveling and the geoid model. Colombia and Chile present a random behavior. In Brazil, $\Delta\zeta$ has a particular circumstance, because the country has two LVDs. While 95% of the GNSS/leveling stations are connected to the Imbituba tide gauge, 5% are related to the Santana tide gauge. This is because it has not yet been possible to cross the Amazon River and the Amazon forest with a spirit level to connect all the Brazilian states in a unique network.

4.1.3 Computation of potential values at IHRF stations based on recent geoid models

In South America, 17 stations (Fig. 10) are planned for the first realization of IHRF. We computed the geopotential numbers of these stations using the geoid undulation values of SAM_GEOID2021 and SAM_GEOID2023. To keep the article concise, we will not elaborate on the formulation here; however, it can be consulted in Sánchez et al. (2021). The stations in the Andes region presented the highest difference in the comparison of both geoid models. The differences between RIOP and MEDE stations are 9.89 and 6.39 m^2s^{-2}, respectively. AREQ, UNSA,

Table 6 Summary of absolute statistical analysis of standard deviation, minimum and maximum values (in meters) after the correction surface

	EGM2008 (2190)	EIGEN6C4 (2190)	XGM2019e (2190)	SGG– UGM– 2 (2190)	SAM_GEOID2023 SAM_QGEOID2023
Argentina (2922 points)					
STD	0.27	0.25	0.24	0.25	0.23
Min	– 1.28	– 1.06	– 1	– 1.04	– 1.22
Max	1.77	1.8	2	1.79	1.84
Brazil – Imbituba (1271 points)					
STD	0.22	0.15	0.13	0.16	0.12
Min	– 0.63	– 1.6	– 0.6	– 1.74	– 0.71
Max	2.71	0.71	0.69	0.98	0.63
Brazil – Santana (66 points)					
STD	0.18	0.09	0.08	0.1	0.06
Min	– 0.49	– 0.22	– 0.18	– 0.24	– 0.11
Max	0.37	0.22	0.2	0.21	0.17
Chile– Arica (27 points)					
STD	0.17	0.22	0.17	0.17	0.09
Min	0.02	– 0.22	0.34	– 0.11	– 0.26
Max	0.59	0.51	– 0.22	0.46	0.13
Chile– Valparaíso (127 points)					
STD	0.18	0.16	0.16	0.16	0.15
Min	– 0.28	– 0.29	– 0.31	– 0.3	– 0.34
Max	0.51	0.51	0.59	0.5	0.54
Chile– Talcahuano (118 points)					
STD	0.16	0.14	0.15	0.15	0.15
Min	– 0.61	– 0.6	– 0.63	– 0.63	– 0.64
Max	0.6	0.65	0.72	0.65	0.73
Chile– Puerto Montt (84 points)					
STD	0.26	0.22	0.25	0.21	0.2
Min	– 0.28	– 0.27	– 0.34	– 0.24	– 0.49
Max	1.11	0.96	1.04	0.95	0.53
Colombia (5837 points)					
STD	0.4	0.36	0.35	0.35	0.32
Min	– 2.75	– 2.43	– 2.48	– 2.39	– 2.38
Max	2.02	1.68	1.97	1.73	1.62
Ecuador (692 points)					
STD	0.46	0.43	0.42	0.42	0.36
Min	– 1.31	– 1.21	– 1.41	– 1.27	– 1.4
Max	1.72	2.1	2	2.08	1.26
Uruguay (44 points)					
STD	0.09	0.08	0.08	0.07	0.07
Min	– 0.18	– 0.18	– 0.18	– 0.16	– 0.13
Max	0.17	0.16	0.14	0.17	0.2
Venezuela (180 points)					
STD	0.2	0.16	0.16	0.16	0.15
Min	– 0.51	– 0.46	– 0.46	– 0.47	– 0.43
Max	0.66	0.53	0.53	0.56	0.61

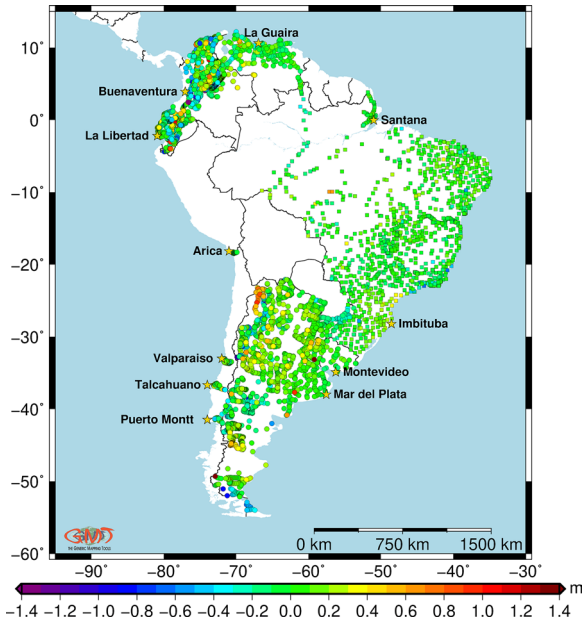


Fig. 8 Absolute statistical analysis between GNSS/leveling data and SAM_GEOID2023 (circle dots) and SAM_QGEOID2023 (square dots). The stars represent the LVD of each country

and SANT stations also presented differences above 2.0 m^2s^{-2}.

Two aspects can explain the differences observed in the Andes region. The first one concerns the Downward Continuation (DC) process. In the SAM_GEOID2021 model, DC was not applied. According to Sideris and Forsberg (1991) and Martinec (1996), the downward continuation can improve the geoid models in mountain areas. The second aspect is about the gravity anomaly grid build. In the SAM_GEOID2021, the punctual terrestrial free-air gravity anomalies were transformed into a 5' resolution grid. Cells lacking information over land and sea were filled with gravity anomaly from XGM2019e and DTU17GRA in 5' grids, respectively. Subsequently, a 5' grid of the entire area was generated, with the coordinates at the center of the grid, using the SURFACE program from the GMT computational package. This program grids table data using adjustable tension continuous curvature splines (Wessel et al. 2013). In contrast, the SAM_GEOID2023 model involved converting each gravity point into a complete Bouguer gravity anomaly. The complete Bouguer gravity anomalies of survey gravity and XGM2019e gravity anomaly (covering the gap area) were interpolated into a grid of 1'×1' using the GEOGRID program of GRAVSOFT computational package (Collocation (kriging) option). Subsequently, the grid was averaged into a 5'×5' grid (BLOCKMEAN program–GMT computational package). The SHGeo computational package was then used to obtain the mean free-air gravity anomalies from the mean complete Bouguer gravity anomalies. In addition, the ocean areas were infilled with grav_32.1 data in 5' grids. It is worth noting that the distribution of gravity coverage around the aforementioned stations is not well-distributed, as gathering terrestrial gravity measurements in the Andes region poses challenges, which can impact the construction of the gravity grid.

4.2 *Local vertical datum parameters and W_0*

The vertical datum offset for each GNSS/leveling station was estimated by applying Eq. (17). The results were then standardized in a second step. Table 7 presents a comparison of the residuals obtained from both computations with the original and standardized input data. It is worth noting that the homogenized data were used in the geoid and quasi-geoid validation (Sect. 4.1). Colombia has a substantial amount of data (5921 points). However, 84 stations were removed in the homogenization process. It was detected that some of those stations presented unprecise information, which resulted in a residual of 61.65 m (max) and − 99.34 m (min).

The impact of systematic distortions and tilt errors on determining vertical data offsets is analyzed using Eq. (17). Table 8 presents the numerical results of two scenarios, referred to as "non-tilts" and "with-tilts" for clarity. The table shows that accounting for systematic tilt and distortions significantly improved the accuracy of the vertical datum parameter for Argentina (Mar del Plata) by 68% (from 0.27 m to 0.08 m) in terms of standard deviation. In Brazil, Imbituba LVD was enhanced by 28% (from 0.18 m to 0.13 m), while Santana LVD by 49% (from 0.09 m to 0.05 m). The most significant enhancement in Chile was Talcahuano, with 82% (from 0.16 m to 0.03 m). In Colombia, the improvement was 72%. In Ecuador, the improvement was 64% while in Uruguay, 66% and in Venezuela, 62%.

The reference potential in each LVD (Table 9) was computed based on Eq. (4) concerning the global reference level $W_0 = 62{,}636{,}853.4\ m^2s^{-2}$. The offset, in terms of potential and meters units and the tilt in east–west and north–south directions, was estimated considering the systematic errors and distortions in the leveling network by applying the least squares adjustment [Eq. (18)]. The number of GNSS/leveling stations impacts the vertical datum parameter standard deviation. In Argentina, Brazil (Imbituba), and Colombia, 2922, 1271, and 5837 stations were used in the computation, respectively. The standard deviation was 0.03 m, 0.02 m, and 0.03 m, respectively. On the other hand, in Chile (Arica), only 28 stations were used in the computation and in Uruguay (44 stations); the standard deviation was 0.25 m, and 0.26 m, respectively.

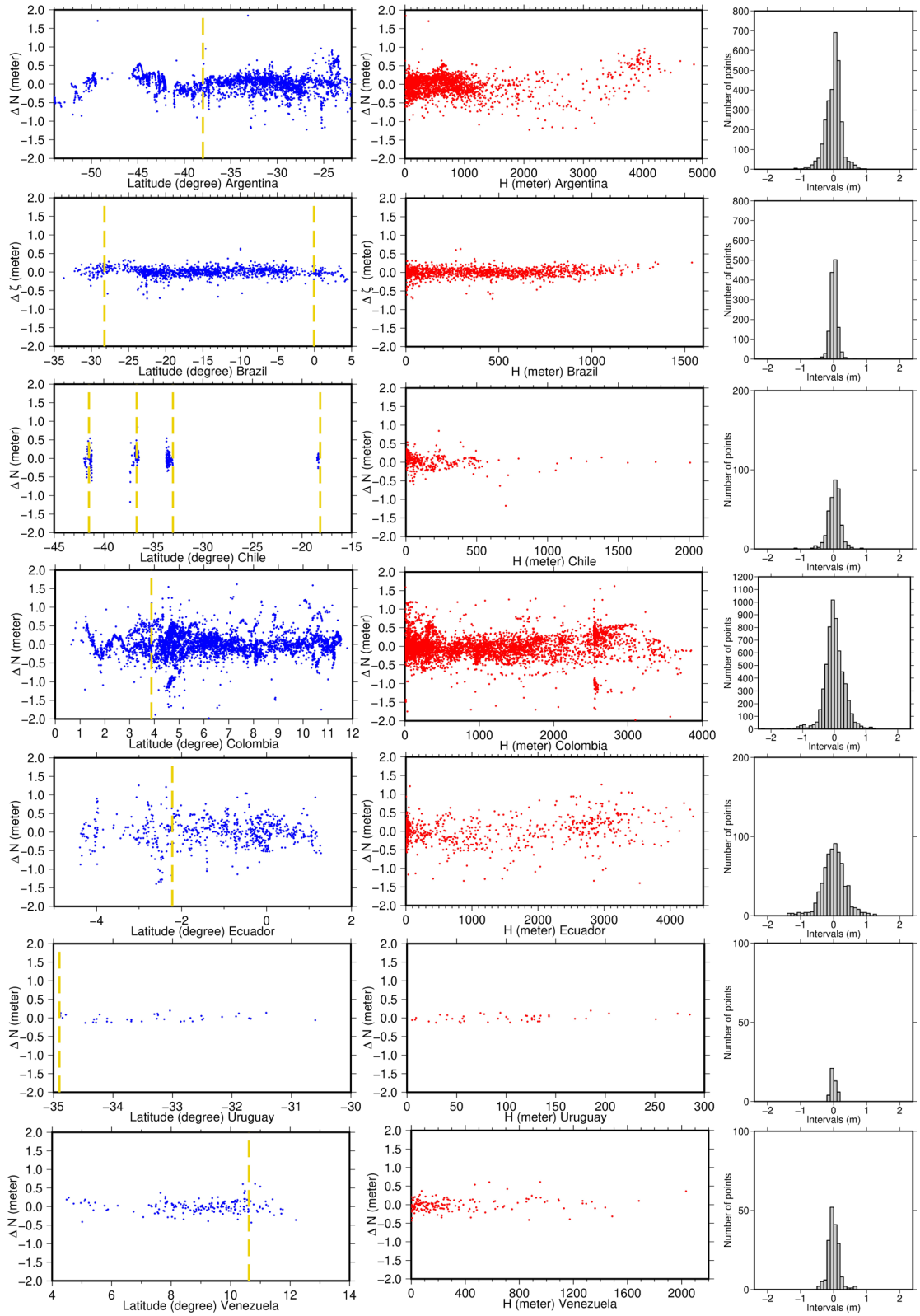


Fig. 9 Difference of ΔN and $\Delta\zeta$ as a function of the latitude (blue dots on the left side). At the center, the difference of ΔN and $\Delta\zeta$ as a function of the height. Yellow dashed lines are the locations of each local vertical data. On the right side, the histograms

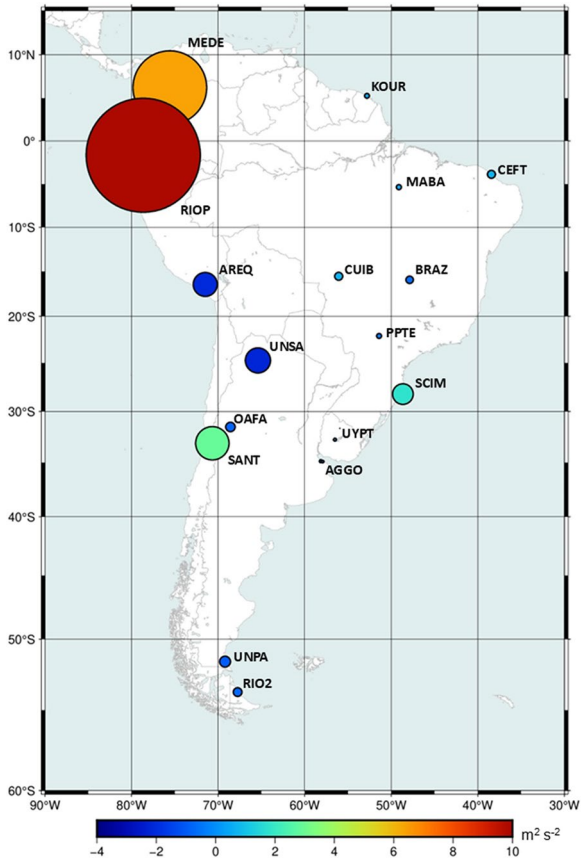


Fig. 10 Difference of geopotential numbers computed from SAM_GEOID2021 and SAM_GEOID2023 models at IHRF stations

It is essential to consider the quality of the leveling operations, especially in rugged topographies, like the Chilean territory due to the Andes Mountains.

The results presented in this paper were compared to the values obtained in South America by Sánchez and Sideris (2017) and in Argentina by Gómez et al. (2024). Sánchez and Sideris (2017) used the GBVP approach, while Gómez et al. (2024) presented two approaches; the first based on GBVP and the second combines geopotential numbers obtained with leveling and gravity with geopotential numbers derived from a quasigeoid model.

In Argentina, the vertical datum parameter found in this paper was 0.71 ± 0.03 m (Fig. 11). Sánchez and Sideris (2017) found 0.66 ± 0.05 m in their study using 48 observations. Gómez et al. (2024) found -0.05 ± 0.18 m and -0.05 ± 0.14 m in both approaches. In Brazil (Imbituba), the difference between our results and those of the authors mentioned above is -0.04 m. We obtained 0.36 ± 0.02 m in this study, while the authors estimated 0.39 ± 0.02 m (with 436 observations). In Chile, Sánchez and Sideris (2017) used three stations in Puerto Montt and two in the other LVDs. The results achieved in their study (Table 8 and Fig. 11) were 0.50 ± 0.24 m (Arica), 0.29 ± 0.18 m (Valparaíso), -0.03 ± 0.18 m (Talcahuano), and 0.30 ± 0.17 m (Puerto Montt). The discrepancies can be attributed to the number of observations used in this study compared to Sánchez and Sideris (2017). Besides that, the authors used an observation equation based on ellipsoidal heights (derived from continuous GNSS station) geopotential numbers derived from leveling, disturbing potential values inferred from national geoid or

Table 7 Comparison of the residuals obtained after estimating the vertical datum offset applying all GNSS/leveling stations in use (left) and the data homogenized in this process (right)

Local Vertical Datum	Residual in (m) using the original data				Residual in (m) using the homogenized data			
	Num. of obs	Mean and standard deviation	Max	Min	Num. of obs	Mean and standard deviation	Max	Min
Argentina	2931	0.71 ± 0.28	2.76	− 1.32	2922	0.72 ± 0.27	2.76	− 0.59
Brazil-Imbituba	1283	0.38 ± 0.19	1.34	− 0.58	1271	0.38 ± 0.18	0.76	− 0.34
Brazil-Santana	67	1.70 ± 0.10	2.11	− 1.46	66	1.69 ± 0.09	1.93	1.46
Chile-Arica	28	0.56 ± 0.26	0.93	− 0.48	27	0.60 ± 0.17	0.93	0.31
Chile-Valparaíso	135	0.71 ± 1.60	10.61	− 4.57	127	0.51 ± 0.20	0.87	0.1
Chile-Talcahuano	119	0.28 ± 0.20	1.01	− 1.02	118	0.29 ± 0.16	1.01	− 0.64
Chile-Puerto Montt	86	0.37 ± 0.75	2.15	− 4.75	84	0.41 ± 0.48	1.29	− 0.67
Colombia-Buenaventura	5921	0.89 ± 2.77	61.65	− 99.34	5837	0.99 ± 0.34	2.58	− 1.32
Ecuador-La Libertad	704	1.05 ± 0.73	5.22	− 11.6	692	1.03 ± 0.39	2.17	− 0.48
Uruguay-Montevideo	44	0.82 ± 0.09	0.99	0.67	44	0.82 ± 0.09	0.99	0.67
Venzuela-La Guaira	190	0.71 ± 0.41	3.07	− 1.22	180	0.69 ± 0.19	1.29	0.2

Table 8 Descriptive statistics for the non-tilt and with-tilt scenarios regarding estimated vertical data offset based on each LVD (in meters)

Country	LVD	Scenario	Mean and STD	Max	Min
Argentina	Mar del Plata	Non-tilts	0.72 ± 0.27	2.76	− 0.59
		With-tilts	0.72 ± 0.08	0.97	0.53
Brazil	Imbituba	Non-tilts	0.38 ± 0.18	0.76	− 0.34
		With-tilts	0.38 ± 0.13	0.58	− 0.12
	Santana	Non-tilts	1.69 ± 0.09	1.93	1.46
		With-tilts	1.69 ± 0.05	1.73	1.54
Chile	Arica	Non-tilts	0.60 ± 0.17	0.93	0.31
		With-tilts	0.60 ± 0.13	0.82	0.4
	Valparaíso	Non-tilts	0.51 ± 0.20	0.87	0.1
		With-tilts	0.51 ± 0.14	0.88	0.3
	Talcahuano	Non-tilts	0.29 ± 0.16	1.09	− 0.64
		With-tilts	0.29 ± 0.03	0.32	0.16
	Puerto Montt	Non-tilts	0.41 ± 0.48	1.29	− 0.67
		With-tilts	0.41 ± 0.42	0.85	− 0.59
Colombia	Buenaventura	Non-tilts	0.99 ± 0.34	2.58	− 1.32
		With-tilts	0.99 ± 0.10	1.24	0.55
Ecuador	La Libertad	Non-tilts	1.03 ± 0.39	2.17	− 0.48
		With-tilts	1.03 ± 0.14	1.31	0.71
Uruguay	Montevideo	Non-tilts	0.82 ± 0.09	0.99	0.67
		With-tilts	0.82 ± 0.03	0.88	0.75
Venezuela	La Guaria	Non-tilts	0.69 ± 0.19	1.29	0.2
		With-tilts	0.69 ± 0.07	0.88	0.56

quasi-geoid models of EGM2008 (where nations models are not available). Utilizing a more significant amount of data (good quality) is believed to lead to consistent results. In Colombia, the offset value is 0.99 ± 0.03 m (Table 8 and Fig. 11), while in Sánchez and Sideris (2017) 0.44 ± 0.03 m. In this study, 5837 points were used, while in the study of the mentioned authors, 83 observations. In Ecuador, Uruguay, and Venezuela, the vertical datum parameter differences between this study and Sánchez and Sideris (2017) are 0.28 m, 0.25 m, and 0.17 m. They used 23, 13, and 33 observations, respectively. All the eleven datum parameters computed present positive values (Fig. 11), meaning the local vertical surface is above the reference level (W_0).

Gomez et al. (2015) analyzed the discrepancies between the vertical reference frames of Argentina (AR) and Brazil (BR). In three connection points along the border, they found − 0.98 m (in Santo Tomé—AR and São Borja—BR), − 0.55 m (Alvear—AR and Itaquí—BR), and − 0.53 (in Passo de los Libres—AR and Uruguaiana—BR). This study found − 0.41 m, − 0.40 m, and − 0.40 m, respectively. The distance between the northernmost connection (Santo Tomé—AR and São Borja—BR) and the southernmost connection Passo de los Libres—AR and Uruguaiana—BR) is around 185 km.

The offset values in Argentina and Brazil decrease in the east–west direction (Fig. 12). The offset values in Uruguay and the East coast of Argentina (close to *Río de La Plata*) are near, and the reference potential difference between these two countries is 1.02 m^2s^{-2}. The reference potential difference between Colombia and Ecuador is 0.46 m^2s^{-2}, which indicates that the local reference surfaces are pretty close to each other (Fig. 12). In Brazil, the offset between Santana and Imbituba LVDs (Fig. 11) is 1.33 ± 0.42 m. The value obtained in this study is compatible with the investigations of Montecino and de Freitas (2014), Moreira and Freitas (2016), Santacruz Jaramillo et al. (2019a), Santacruz Jaramillo and Freitas (2021).

Table 9 Vertical datum parameters and potential values in LVD concerning the global reference level $W_0 = 62{,}636{,}853.4\ m^2s^{-2}$

Country	LVD	Reference Potential (m^2s^{-2})	Potential Parameter (m^2s^{-2})	Vertical Parameter (m)	East–West Tilt (m/°)	North–South Tilt (m/°)
Argentina	Mar del Plata	62,636,846.37 ± 0.27	6.95 ± 0.27	0.71 ± 0.03	0.02	− 0.01
Brazil	Imbituba	62,636,849.87 ± 0.22	3.50 ± 0.22	0.36 ± 0.02	0.03	− 0.01
	Santana	62,636,836.88 ± 4.11	16.52 ± 4.11	1.69 ± 0.42	0.1	− 0.04
Chile	Arica	62,636,847.57 ± 2.44	5.82 ± 2.44	0.59 ± 0.25	3.04	− 0.84
	Valparaíso	62,636,848.42 ± 0.94	4.99 ± 0.94	0.51 ± 0.10	− 0.52	0.71
	Talcahuano	62,636,850.54 ± 0.56	2.87 ± 0.56	0.29 ± 0.06	− 0.27	0.03
	Puerto Montt	62,636,849.40 ± 1.29	4.04 ± 1.29	0.41 ± 0.13	− 1.8	1.28
Colombia	Buenaventura	62,636,843.74 ± 0.26	9.70 ± 0.26	0.99 ± 0.03	− 0.17	− 0.02
Ecuador	La Libertad	62,636,843.28 ± 0.79	10.08 ± 0.79	1.03 ± 0.08	0.25	0.08
Uruguay	Montevideo	62,636,845.35 ± 2.52	8.03 ± 2.52	0.82 ± 0.26	− 0.05	− 0.01
Venezuela	La Guaria	62,636,846.62 ± 1.04	6.80 ± 1.04	0.69 ± 0.11	− 0.02	− 0.05

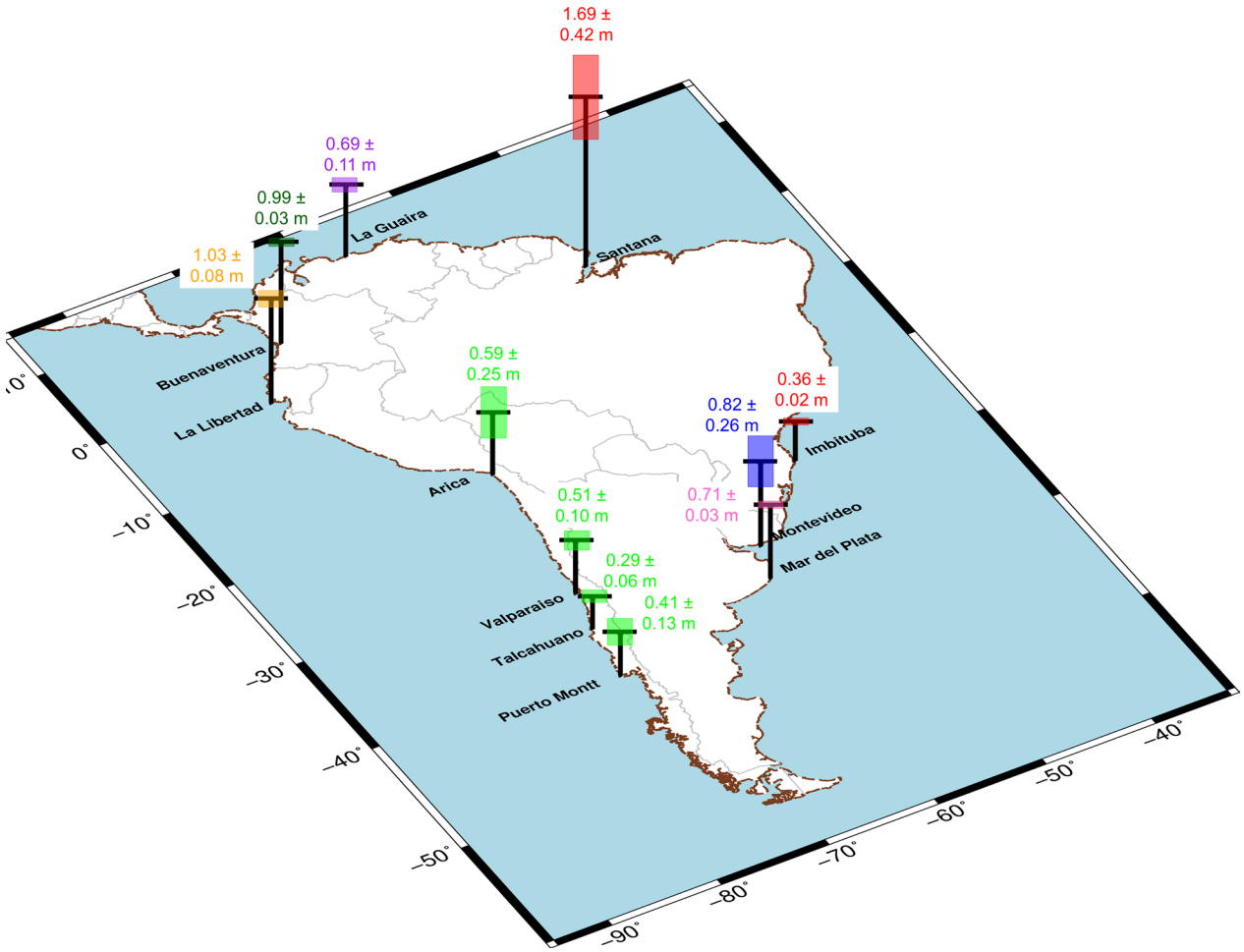


Fig. 11 Vertical datum parameter and standard deviation values (in meters) in each LVD related to $W_0 = 62{,}636{,}853.4\ m^2s^{-2}$ (brown dashed line). Black bars represent the separation between the LVD and the reference level. The color bars represent the standard deviation of each offset

5 Conclusions

The global unification of the vertical datum has been a long-standing issue in geodesy. From 2010 to 2023, the unified height system was part of the GGOS focus area, which aimed to establish a Global Unified Height reference system. This study estimated geopotential values and local vertical datum parameters using a geoid (quasi-geoid) model in seven South American countries.

Initially, we computed a geoid model for the entire study area, taking into account several important factors: (1) gravity data (collected from different countries using distinct types of equipment in different epochs) to calculate the gravity anomalies, represented significant efforts in terms of the treatment and homogenization data. Besides that, the gravity gaps in certain areas compromise the model's accuracy and (2) the topography in South America is divided into mountains and highlands, river basins, and coastal plains. Modeling the topographic effects using a global DEM, particularly in the Andes region, was a significant undertaking. We believe that the quality of the models can be affected by the two factors mentioned above. Therefore, we suggest that each country compute a model specifically for its region. Sometimes, national agencies may have access to new gravity data and/or local or regional DEM, which can enhance the model's quality.

We validated GNSS/leveling data using SAM_GEOID2023, SAM_QGEOID2023, and GGMs. Applying the correction surface to the GNSS/leveling data via the four-parameter model results in a slight standard deviation. In Brazil–Imbituba, the standard deviation decreased 33% while in Chile–Arica, 48%, and Chile–Puerto Montt, 42%.

It is important to consider the homogenization of geodetic parameters, especially regarding the reference system and tidal concepts when working with the geoid (quasi-geoid) and calculating potential values. To ensure

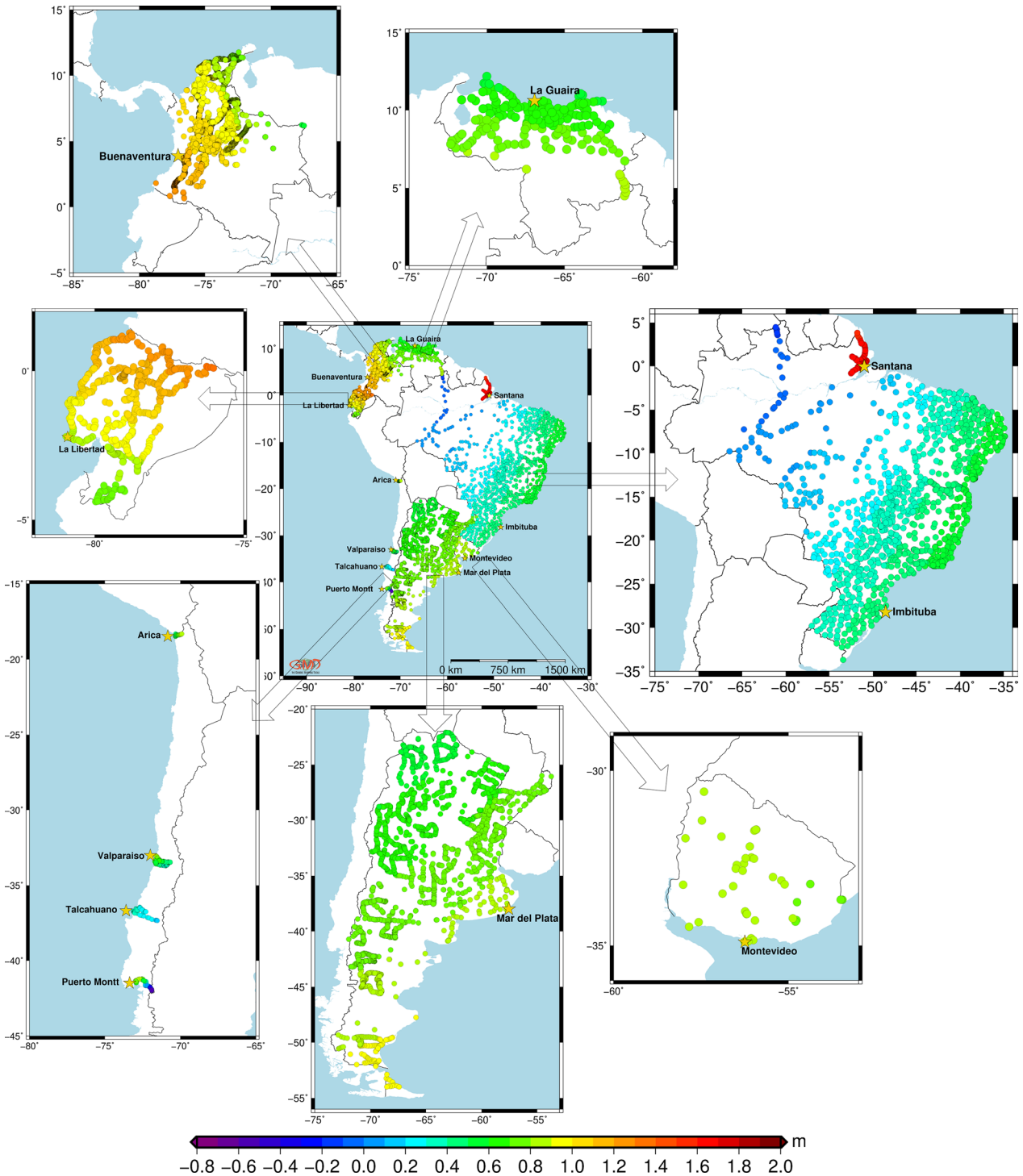


Fig. 12 Vertical datum offsets related to the GNSS/leveling stations in South America

consistency, we performed the proper transformation during the computation process.

Regarding computing the potential value, the homogenization data to remove unprecise GNSS/leveling data was fundamental. In addition, the data adjustment by the parametric model minimized both random and systematic errors, and the results were more consistent than the data not adjusted. In South America, the vertical datum

parameter, considering the seven studied countries, varied from 0.29 m (in Chile) to 1.69 m (in Brazil). For countries with two local vertical data, the offset between Imbituba and Santana was estimated to be 1.33 ± 0.42 m. This means the Santana vertical datum is 1.33 m higher than the Imbituba vertical datum.

Finally, we computed potential reference values and the vertical datum parameters in eleven LVDs by combining geoid (quasi-geoid) model with GNSS/leveling stations. The standard deviation found is related to the quality and amount of GNSS/leveling data utilized in the method. To improve the reliability of the W_0^{LVD} and the δH, we recommend to the countries to ensure good quality distribution data. We can connect local and global height systems using vertical datum parameters, guaranteeing consistency in the global vertical datum system. We also recommend that countries open their data repository regarding GNSS/leveling data, which will allow future studies on topics such as those presented in this paper.

Abbreviations

MSL	Mean sea level
LVD	Local vertical datum
IUGG	International Union of Geodesy and Geophysics
IAG	International association of geodesy
IHRS	International height reference system
IHRF	International height reference frame
GBVP	Geodetic boundary value problem
GGMs	Global geopotential models
DEM	Digital elevation model
RCR	Remove–compute–restore
EPUSP	Polytechnic School of the University of São Paulo
LAGEOS	Laser geodynamics satellite
GRACE	Gravity recovery and climate experiment
GOCE	Gravity field and steady-state ocean circulation explorer
SRTM	Shuttle radar topography mission
ASTER	Advanced spaceborne thermal emission and reflection radiometer
IGN-AR	Instituto Geográfico Nacional de Argentina
IBGE	Instituto Brasileiro de Geografia e Estatística
IGM	Chile Instituto Geográfico Militar de Chile
IGAC	Instituto Geográfico Agustín Codazzi
IGM	Ecuador Instituto Geográfico Militar de Ecuador
IGM	Uruguay Instituto Geográfico Militar de Uruguay
IGVSB	Instituto Geográfico de Venezuela Simón Bolívar
SIRGAS	Geodetic Reference System for the Americas
RTM	Residual terrain model
GRS80	Geodetic Reference Ellipsoid 1980
ICGEM	International Centre for Global Earth Models
DTE	Direct topographical effect
DAE	Direct atmospheric effect
SITE	Secondary indirect topographical effect
G–QG	Geoid–quasigeoid correction
D	Downward continuation
PITE	Primary indirect topographic effect
PIAE	Primary indirect atmospheric effect
SAM_GEOID2023	South AMerican GEOID 2023

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Author contributions
All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Gabriel do Nascimento Guimarães, Ana Cristina Oliveira Cancoro de Matos and Flavio Guilherme Vaz de Almeida. Denizar Blitzkow performed the project supervision. Gabriel do Nascimento Guimarães wrote the first draft of the manuscript, and all authors commented on previous versions. All authors read and approved the final manuscript.

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Availability of data and materials
The data sets used and/or analyzed during the current study are available from the corresponding author on reasonable request. The models SAM_GEOID2023 and SAM_QGEOID2023 are available on the International Service for Geoid (ISG) website for download.

Declarations

Competing interests
The authors declare no competing interests. The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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